AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

Test Report No. 2023 – 0019

Agricultural Machinery – Furrower
FT 4 - BOTTOM FURROWER

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Test Report Number	2023 – 0019
Date of Issuance	March 7, 2023
Valid Until	March 7, 2028
Agricultural Machinery	Furrower
Type	Tractor-mounted, Four shank, General-purpose bottom
Brand	FT
Model	4 - BOTTOM FURROWER
Test Requested by	FORD TRACTOR PHILS. INC. 1384 Jose Abad Santos, Tondo, Manila

SUMMARY

Performance Criteria	Requirement as per PAES 134:2004	AMTEC Test
Depth of Furrow, mm, maximum	ND[a]	230 ± 17.69
Width of Furrowing, mm, maximum	ND[a]	2691 ± 171.56

[a] Not specified by the requesting party.
Note: ND – No Data

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	FORD TRACTOR PHILS. INC.
4. Four-wheel tractor used	33.1 kW YTO MF454
5. Number of operators	One
6. Handling and stability when machine is working	The furrower was stable and maintained a smooth operation while it is hitched to the tractor. There were no issues encountered when maneuvering at corners.
7. Adjustments of parts	Depth of furrow and furrow spacing can be adjusted by moving the hitch links and standard brackets, respectively.
8. Straightness of furrows	Straightness of furrows was dependent on the operator.
9. Evenness of spacing between furrows	675 mm ± 42.84 mm
10. Uniformity of depth and width	Depth of cut: 230 mm ± 17.69 mm Width of cut: 2691 mm ± 171.56 mm
11. Durability of parts (based on wear of soil-working parts, visible deformation, etc)	No deformations were observed during the test operation.
12. Safety features	None
13. Failures or abnormalities that may be observed on the machine or its components during and after the operation	None
14. The size of the furrower bottom shall be determined by measuring the horizontal distance from the left to the right wings of share.	The size of the furrower bottom is 475 mm.
15. The number of furrower bottoms shall depend on crop's row spacing and the tractor's wheel tread adjustment.	The furrower has four shanks. The manufacturer's recommended spacing for corn was 70 mm.

OBSERVATIONS (Cont'n)

16. The height of the standards shall vary according to the crop requirement but shall not exceed 500 mm	The height of the standards is 450 mm
17. The length of the toolbar shall vary according to crop's row spacing and the tractor's wheel tread adjustment.	The length of the tool bar is 2450 mm.
18. The number of toolbars may be one or two depending on the manufacturer's design.	The furrower has one tool bar.
19. Cast iron and/or mild steel shall be used in the manufacture of the moldboard, standard and toolbar.	Steel was used for the manufacture of moldboard, standard, and toolbar.
20. Carbon steel with at least 80% carbon content (e.g. AISI 1080) or alloy steel with at least 0.0005% boron content shall be used in the manufacture of share.	The material used for the share was not verified.
21. The hitch of the furrower shall be compatible with the three-point linkage of the four-wheel tractor as specified in PAES 118.	Not verified
22. The furrower shall be easy to hitch to and unhitch from the tractor as well as adjust the spacing between furrows.	Hitching and unhitching were easy. Adjusting the furrower spacing can be done by loosening the bolts and nuts used to attach the shanks to the toolbars.
23. The furrower shall be free from manufacturing defects such as sharp edges and surfaces that may be detrimental to the operator.	The furrower was free from manufacturing defects during the test operation.
24. Except for furrower bottoms, other uncoated metallic surfaces shall be free from rust and shall be painted properly.	All parts were painted except for the furrower bottom.
25. Other remarks	No markings or labelling provided on the furrower.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	January 20, 2023
Location of Test	Luisita Access Road, Tarlac City, Tarlac 15°26'5.93" N, 120°39'48.13" E
Items Tested	Operator's Manual, Depth of Furrow, Width of Furrow, Field Capacity, Field Efficiency, Noise Level, and Fuel Consumption

METHODS OF TEST

The furrower was tested based on the PAES 135:2004 – Agricultural Machinery – Furrower – Methods of Test.

DESCRIPTION OF THE MACHINE

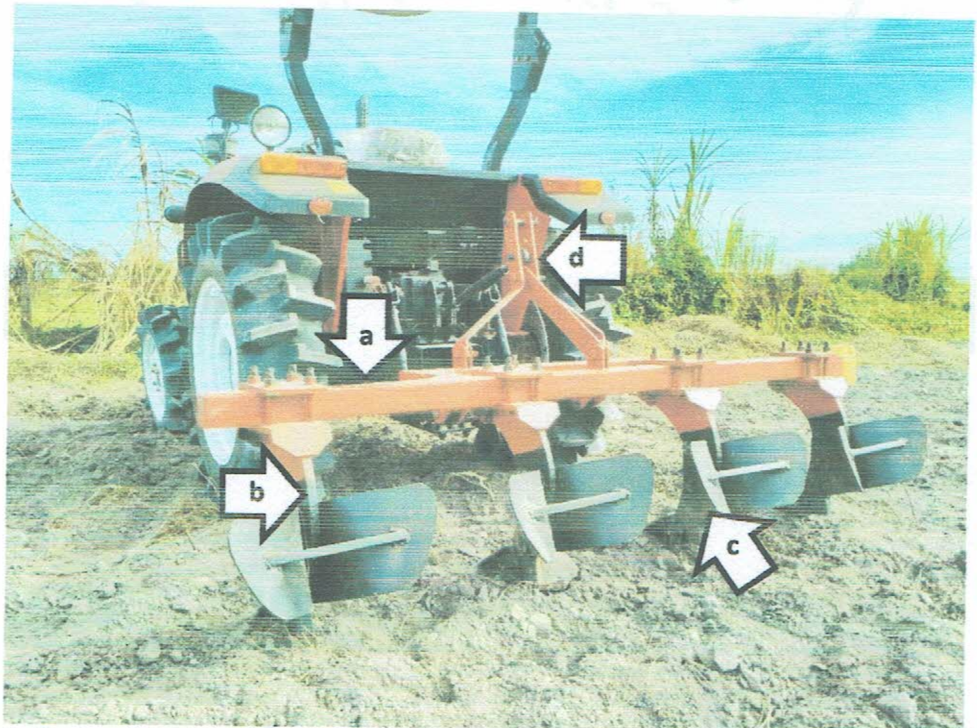

Figure 1. Parts of the FT 4 - BOTTOM FURROWER Furrower hitched to 33.1 kW YTO MF454 include the following: (a) Toolbar, (b) Standard, (c) Moldboard, and (d) Upper-link hitch point.

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification
1	Overall dimensions and weight		
1.1	Length, mm	2530	770
1.2	Width, mm	850	2525
1.3	Height, mm	1240	1100
1.4	Weight, kg	ND	NM
2	Furrower bottom		
2.1	Quantity	4	4
2.2	Size, L × W, mm	ND	2450 × 85
2.3	Material	Leafspring	ND
3	Standard		
3.1	Dimension, H × W × t, mm	600 × 250 × 30	450 × 210 × 24.91
3.2	Material	MS plate	ND
4	Toolbar		
4.1	Dimension, L × t, mm	2210 × 127	2450 × 85
4.2	Material	Mild steel angle bar	ND
5	Features		
5.1	Recommended furrow spacing, mm	ND	700[b]
5.2	Type of crops	ND	Corn[b]
6	Operating width, mm	ND	2525

[a] The data were obtained from the operator's manual and specifications sheet provided by the requesting party.
[b] Specified by the requesting party during field test
Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data
1	Test Conditions	
1.1	Dimensions of field, m	
1.1.1	Length	46.0
1.1.2	Width	23.0
1.2	Area, m^2	1058.00
1.3	Soil type	Sandy loam
1.4	Soil moisture content, %	9.98
1.5	Last crop planted	Mungbean
1.6	Field condition before operation	Harrowed
2	Field Performance	
2.1	Type of field operation	Secondary tillage - Furrowing
2.2	Tractor's gearshift setting	4^{th} gear – Low
2.3	Traveling speed, kph	5.52
2.4	Average depth of furrow, mm	230 ± 17.69
2.5	Average width of furrowing, mm	2691 ± 171.56
2.6	Actual field capacity	
2.6.1	ha/h	1.18
2.6.2	h/ha	0.853
2.7	Theoretical field capacity	
2.7.1	ha/h	1.40
2.7.2	h/ha	0.721
2.8	Field efficiency, %	84.66
2.9	Noise at operator's ear level, dB(A)	91.0
2.10	Wheel slip, %	NM
2.11	Fuel consumption	
2.11.1	L/h	8.30
2.11.2	L/ha	7.09
2.13	Method of operation	Continuous pattern turn strips at each end

Note: NM – Not Measured

OBSERVATION

Figure 2. Quality of work

Tested by:

RHEY MARC C. ALBALOS
Test Engineer

CHRISTOPHER S. JAPONES
Test Engineer

Verified by:

MARIE JEHOSA B. REYES
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO
AMTEC Director

MAR 10 2023
Date

REMINDER

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